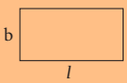
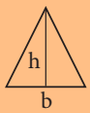
Table 1.3 Area of some regularly shaped objects

S.No.	Plane figure	Diagram	Area
1	Square		side × side $a \times a = a^2$ sq.unit
2	Rectangle		length × breadth $l \times b = lb$ sq.unit
3	Circle		$\pi \times (\text{radius})^2$ $\pi \times r^2 = \pi r^2$ sq.unit
4	Triangle		(1/2) × base × height $1/2 \times b \times h$ sq.unit

Problem 1.2

Find the area of the following regular shaped figures (Take $\pi = 22/7$).

a. A rectangle whose length is 12 m and breadth is 4 m.

b. A circle whose radius is 7 m.

c. A triangle whose base is 6 m and height is 8 m.

Solution

a. Area of rectangle = length × breadth
$= 12 \times 4 = 48 \text{ m}^2$

b. Area of circle $= \pi \times r^2 = (22/7) \times 7 \times 7$
$= 154 \text{ m}^2$

c. Area of triangle = 1/2 × base × height
$= 1/2 \times 6 \times 8 = 24 \text{ m}^2$

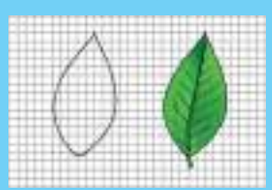
1.2.2 Area of irregularly shaped objects

In our daily life, we encounter many irregularly shaped objects like leaves, maps, stickers of stars or flowers, peacock feather etc. The area of such irregularly shaped objects cannot be calculated using any formula.

How can we find the area of these irregularly shaped objects? We can find the area of these figures with the help of a graph sheet. The following activity shows how to find the area of irregularly shaped plane figures.

ACTIVITY 1

Take a leaf from any one of the trees. Place it on a graph sheet and draw the outline of the leaf with a pencil. Remove the leaf. You can see the outline of the leaf on the graph sheet.



i. Now, count the number of whole squares enclosed within the outline of the leaf. Take it as M.

ii. Then, count the number of squares that are more than half. Take it as N.

iii. Next, count the number of squares which are half of a whole square. Note it as P.

iv. Finally, count the number of squares that are less than half. Let it be Q.

Now, the approximate area of the leaf can be calculated using the following formula.

Approximate area of the leaf
= M + (3/4) N + (1/2) P + (1/4) Q square cm.

Area of the leaf = _____cm^2.

This method can be used to find the area of regularly shaped figures also. In the case of square and rectangle, this method gives the measure area accurately. This method can be used to calculate the area of any irregularly shaped plane figures.

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